

Fuel-Nozzle Technical Reference

Jet-A Pressure-Swirl and Flashing LNG Models

Beginner-Oriented Equations, Algorithms, Source Locations, and Limitations

fuelnozzle project

Source map generated 2026-08-26

Abstract

This report explains the equations and software implementation of the `fuelnozzle` package for a reader with no prior background in fluid mechanics, thermodynamics, atomization, or Python. The package studies two fuel circuits: a Jet-A pressure-swirl atomizer and a liquid-LNG nozzle that may remain liquid, flash inside the nozzle, or flash after discharge. Every major equation is connected to the current Python function and line where it is implemented. The report also separates physical laws from numerical approximations and empirical calibration. It should be read as documentation for a reduced-order screening tool, not as a hardware certification method.

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1 How to Read This Report

A physical equation describes a relationship among measurable quantities. A software implementation turns that relationship into arithmetic and decision logic. The same equation can be implemented correctly while still being unsuitable for a particular injector. Therefore, four questions must always be kept separate:

1. **Was the equation entered correctly?** This is software verification.
2. **Does the numerical method converge?** This is numerical verification.
3. **Does the model match experiments?** This is validation.
4. **Is the experiment similar enough to the intended LNG/Jet-A design?** This is the applicability question.

Source references have the form `src/fuelnozzle/lng.py:256`. They refer to the source state on 2026-08-26. Function names are stable references; line numbers will move when source files are edited.

Engineering caution. The current code is a steady, zero/one-dimensional engineering screen. It does not resolve the three-dimensional vortex, individual vapor bubbles, primary breakup, combustion chemistry, flame stability, or structural integrity.

2 Basic Vocabulary for a New Reader

2.1 Pressure, temperature, and flow

Pressure P (Pa)

Force divided by area. One atmosphere is approximately 101325 Pa. All code inputs are absolute pressures unless an external data source explicitly states otherwise.

Temperature T (K)

Thermodynamic temperature. To convert Celsius to kelvin, add 273.15.

Mass flow $\dot{m}$ (kg/s)

Fuel mass passing a location each second.

Area A (m²)

Flow-passage cross-sectional area.

Density ρ (kg/m³)

Mass per unit volume.

Velocity u (m/s)

Speed of the fluid.

Mass flux G (kg/(m² s))

Mass flow per area:

$$G = \frac{\dot{m}}{A} = \rho u. \quad (1)$$

Mass flux is useful because a nozzle with twice the area carries twice the mass flow when G is unchanged.

2.2 Liquid, vapor, saturation, and flashing

A pure substance can exist as a liquid or vapor. At a given pressure below the critical point, there is a saturation temperature. A liquid colder than saturation is *subcooled*. A liquid at saturation can begin boiling without further heating.

LNG is a mixture, so it does not have one boiling temperature. At a given pressure:

- the **bubble temperature** is where the first vapor bubble can appear;
- the **dew temperature** is where the last liquid droplet disappears.

Flashing is rapid vapor production caused by a pressure reduction. A liquid can be thermodynamically able to flash but remain temporarily liquid because bubble nucleation and growth require finite time. That temporary state is called *metastable*.

2.3 Enthalpy, entropy, and quality

Specific enthalpy h (J/kg)

An energy-like state property useful for flow energy balances. In an adiabatic throttling process, h is often approximately constant.

Specific entropy s (J/(kg K))

A thermodynamic state property. An ideal adiabatic, reversible acceleration follows constant entropy.

Vapor quality x

Vapor mass divided by total liquid-plus-vapor mass. Thus $x = 0$ is all liquid and $x = 1$ is all vapor.

2.4 Choking

A flow is choked when reducing downstream pressure no longer increases mass flow. For a flashing liquid, this may occur because vapor production makes the fluid highly compressible. The current Tier 1 code calls a case choked when the maximum sampled mass flux occurs at a pressure measurably above the outlet pressure.

2.5 Atomization terms

Pressure-swirl atomizer

A nozzle that sends liquid through tangential ports, forming a rotating liquid film and usually a hollow cone.

Air core

The low-density central region inside the rotating liquid annulus.

Film thickness

Radial thickness of liquid at the exit.

Cone angle

Angular width of the spray. This report distinguishes semi-angle from full cone angle. The code reports full cone angle.

SMD or D_{32}

Sauter mean diameter, the diameter of a hypothetical uniform spray with the same volume-to-surface-area ratio as the actual droplets.

D_{10} Arithmetic mean droplet diameter. It is not interchangeable with SMD.

3 Software Architecture and Data Flow

For each flight-stage operating point, the software performs this sequence:

1. Validate user inputs in `OperatingPoint`.
2. Close the pump/feed/nozzle pressure budget.
3. Create or reuse a `CoolProp` state for methane or an LNG composition.
4. Optionally march the LNG through a heated feed line.
5. Run LNG Tier 0, Tier 1, Tier 2, and Tier 3 in order.
6. Independently run the Jet-A pressure-swirl calculation.
7. Gather warnings instead of silently repairing inconsistent inputs.
8. Integrate stage mass flows when all durations are available.

The public study owner is `run_nozzle_study` at `src/fuelnozzle/study.py:75`. It creates one `CoolPropLNGProvider` and reuses it for all operating points.

4 Input Pressure Contract

The user supplies compressor/combustor station values P_3, T_3 , pump discharge pressure, and nozzle pressure drop. The model currently treats P_3 as the receiving pressure immediately downstream of each fuel nozzle.

The required nozzle inlet pressure is

$$P_{n,\text{in}} = P_3 + \Delta P_n. \quad (2)$$

The pressure remaining for pipes, bends, valves, and manifolds is

$$\Delta P_{\text{feed,available}} = P_{\text{pump,out}} - P_{n,\text{in}}. \quad (3)$$

Equations (2)–(3) are implemented by `fuel_pressure_budget` at `src/fuelnozzle/operating.py:55`, specifically lines 62–63.

If Equation (3) is negative, the point is hydraulically infeasible. The code returns an error-level warning but does not alter P_3 or any pressure input.

Engineering caution. Real gas-turbine notation often uses P_3 for compressor discharge, while pressure at the fuel nozzle can differ due to diffuser and liner losses. The current interface does not separate these locations. Supply the pressure appropriate to the nozzle receiving environment, or extend the data model.

5 Thermodynamic Property Model

5.1 LNG composition

The user supplies nonnegative component mole fractions z_i . The input validator normalizes them:

$$z_i^{\text{normalized}} = \frac{z_i}{\sum_j z_j}. \quad (4)$$

This occurs in `LNGComposition.validate_and_normalize` at `src/fuelnozzle/models.py:37`.

5.2 CoolProp state objects

`CoolPropLNGProvider.__init__` at `src/fuelnozzle/properties.py:29` creates one reusable CoolProp `AbstractState`. Pure methane uses direct CoolProp input pairs. For a mixture, CoolProp supports PT, PQ, and TQ state updates but not every PH or PS flash needed by the nozzle path.

5.3 Mixture PH and PS roots

For a requested pressure P and target enthalpy h^* , the code finds a temperature that satisfies

$$f_h(T) = h(P, T, \mathbf{z}) - h^* = 0. \quad (5)$$

For target entropy s^* , it solves

$$f_s(T) = s(P, T, \mathbf{z}) - s^* = 0. \quad (6)$$

The common implementation is `_mixture_target_state` at `src/fuelnozzle/properties.py:152`. The residual is defined at line 160, and SciPy’s bracketed Brent method is called at line 210. The previous station’s temperature is used to find a nearby bracket; a wider scan is used only when local continuation fails.

This approach is slower than a direct flash, but a bracketed root is robust and does not require inventing mixture thermodynamics outside CoolProp.

5.4 Molar quality to mass quality

CoolProp’s mixture quality is molar. The hydraulics require mass quality. Let Q be molar vapor fraction, M_v vapor-phase molar mass, and M_l liquid-phase molar mass. Vapor and liquid masses for an arbitrary one-mole total basis are

$$m_v = QM_v, \quad (7)$$

$$m_l = (1 - Q)M_l. \quad (8)$$

The mass quality is

$$x = \frac{QM_v}{QM_v + (1 - Q)M_l}. \quad (9)$$

Implemented by `_mass_quality` at `src/fuelnozzle/properties.py:370`, with phase molar masses at lines 375–390 and Equation (9) at lines 391–393.

5.5 Mixture transport fallback

When CoolProp has no native LNG-mixture viscosity or conductivity, the provider uses component values. Viscosity uses logarithmic mole mixing:

$$\mu_{\text{mix}} = \exp \left(\sum_i \hat{z}_i \ln \mu_i \right), \quad \hat{z}_i = \frac{z_i}{\sum_{j \in \mathcal{A}} z_j}, \quad (10)$$

where $\mathcal{A}$ is the set of components for which a value is available. Conductivity and surface tension use linear mole mixing:

$$k_{\text{mix}} = \sum_i \hat{z}_i k_i, \quad (11)$$

$$\sigma_{\text{mix}} = \sum_i \hat{z}_i \sigma_i. \quad (12)$$

Implemented by `_component_transport_fallback` at `src/fuelnozzle/properties.py:281`; Equations (10)–(12) are at lines 345–367.

If a pure component cannot be forced into a liquid PT state, its saturated-liquid transport at the same temperature is attempted. Any omitted component is named in a warning.

Engineering caution. These transport mixing rules are engineering fallbacks, not a validated LNG transport model. Density, enthalpy, entropy, and equilibrium still come from the mixture EOS; only missing transport quantities are approximated.

6 Cryogenic LNG Feed Line

6.1 Segmentation

The feed line is divided into equal axial segments. At each segment, the solver:

1. reconstructs the state from local P and h ;
2. estimates heat ingress;
3. selects single- or two-phase pressure loss;
4. subtracts pressure loss;
5. adds heat to enthalpy.

The owner is `solve_lng_feed_line` at `src/fuelnozzle/feed.py:63`.

6.2 Pipe area and mass flux

For inner diameter D ,

$$A_p = \frac{\pi D^2}{4}, \quad G = \frac{\dot{m}}{A_p}. \quad (13)$$

Implemented at `src/fuelnozzle/feed.py:94–95`.

6.3 Segment energy update

If heat leak per length is q' and segment length is Δz , segment heat is $q'\Delta z$. Dividing by mass flow gives the enthalpy rise:

$$h_{j+1} = h_j + \frac{q'_j \Delta z}{\dot{m}}. \quad (14)$$

Implemented at `src/fuelnozzle/feed.py:148`.

6.4 Internal heat transfer

The standard dimensionless groups are

$$\text{Re} = \frac{GD}{\mu}, \quad (15)$$

$$\text{Pr} = \frac{c_p \mu}{k}, \quad (16)$$

$$h_i = \frac{\text{Nu} k}{D}. \quad (17)$$

ChEDL `ht` selects the internal-flow Nusselt correlation, and cylindrical conduction combines internal convection, wall, insulation, and external convection. The implementation is `_heat_leak_per_length` at `src/fuelnozzle/feed.py:230`, with the quantities at lines 244–267. A measured heat leak in W/m bypasses this correlation chain.

6.5 Single-phase pressure loss

The dynamic pressure written in terms of mass flux is

$$q_d = \frac{G^2}{2\rho}. \quad (18)$$

For Darcy friction factor f , distributed minor-loss coefficient K , segment length Δz , and diameter D ,

$$\Delta P_{f+K} = \left(f \frac{\Delta z}{D} + K \right) \frac{G^2}{2\rho}. \quad (19)$$

Static head is

$$\Delta P_g = \rho g \Delta y. \quad (20)$$

Implemented by `_liquid_segment_pressure_drop` at `src/fuelnozzle/feed.py:270`, with Equations (18)–(20) at lines 283–294.

6.6 Two-phase pressure loss

ChEDL `two_phase_dP` supplies frictional loss. The current selection is `Kim_Mudawar` when $D \leq 6.22$ mm and `Chisholm` for larger lines. Static head uses homogeneous mixture density

$$\rho_m = \left(\frac{1-x}{\rho_l} + \frac{x}{\rho_v} \right)^{-1}. \quad (21)$$

Implemented by `_two_phase_segment_pressure_drop` at `src/fuelnozzle/feed.py:297`; correlation call and density occur at lines 315–333.

The current feed model does not add a separate two-phase acceleration pressure-drop term. This is a known limitation when quality changes rapidly.

7 LNG Tier 0: Is Flashing Thermodynamically Possible?

7.1 Subcooling margins

The temperature subcooling margin is

$$\Delta T_{sub} = T_{bubble}(P, \mathbf{z}) - T. \quad (22)$$

The pressure subcooling margin is

$$\Delta P_{sub} = P - P_{bubble}(T, \mathbf{z}). \quad (23)$$

Positive values indicate equilibrium liquid margin. Implemented by `screen_lng_flash` at `src/fuelnozzle/lng.py:151`, specifically lines 192–193.

7.2 Equilibrium exit quality

The nozzle inlet enthalpy is flashed isenthalpically to P_3 . If the result is two-phase, its mass quality is the equilibrium flash fraction. This is a thermodynamic availability calculation, not a bubble-growth prediction.

7.3 Equilibrium onset location

The code assumes pressure changes linearly from nozzle inlet pressure P_{in} to P_3 . If bubble pressure is P_b , the fractional axial position is

$$\xi_{eq} = \frac{P_{in} - P_b}{P_{in} - P_3}. \quad (24)$$

Implemented at `src/fuelnozzle/lng.py:223`. If $\xi_{eq} \geq 0.98$, the screen calls onset an exit event; otherwise it calls it internal.

Engineering caution. Equation (24) is a geometric screen. The code does not solve the internal nozzle pressure profile from momentum and wall friction.

8 LNG Tier 1: SPI and HEM Mass-Flow Bounds

8.1 Single-phase incompressible bound

Bernoulli's equation gives ideal liquid speed $u = \sqrt{2\Delta P/\rho}$. Multiplying by density gives

$$G_{SPI} = \sqrt{2\rho_0\Delta P_n}. \quad (25)$$

Implemented at `src/fuelnozzle/lng.py:302–306`.

8.2 Homogeneous equilibrium path

HEM assumes liquid and vapor share velocity, pressure, and temperature and reach local phase equilibrium instantly. The current path is isentropic:

$$s(P) = s_0. \quad (26)$$

At each sampled pressure, stagnation enthalpy is converted to kinetic energy:

$$u(P) = \sqrt{2 \max(0, h_0 - h(P, s_0))}. \quad (27)$$

Mass flux follows:

$$G_{HEM}(P) = \rho(P, s_0)u(P). \quad (28)$$

Implemented by `solve_lng_equilibrium_flow` at `src/fuelnozzle/lng.py:256`; Equations (27)–(28) are lines 281–283.

8.3 Critical condition and choking decision

The sampled critical mass flux is

$$G^* = \max_{P_3 \leq P \leq P_{in}} G_{HEM}(P), \quad P^* = \arg \max_P G_{HEM}(P). \quad (29)$$

The software reports choking when

$$P^* > P_3 + \epsilon_P P_{in}, \quad (30)$$

where default $\epsilon_P = 0.005$. This logic is at lines 295–300 of `solve_lng_equilibrium_flow`.

This is a numerical maximum criterion, not an explicit sonic-eigenvalue calculation. Grid convergence is therefore a required verification test.

8.4 Area and diameter sizing

For discharge coefficient C_d and selected operating mass flux G_{op} ,

$$A_{req} = \frac{\dot{m}}{C_d G_{op}}. \quad (31)$$

For N equal circular holes, diameter per hole is

$$D_{req} = \sqrt{\frac{4A_{req}}{\pi N}}. \quad (32)$$

Implemented at `src/fuelnozzle/lng.py:310–315`.

The discharge coefficient is user-specified (default 0.80). HEM does not itself predict vena-contracta and wall-loss effects.

9 LNG Tier 2: Finite-Rate Relaxation Screen

9.1 Why a relaxation model is needed

A short nozzle may not allow enough time for equilibrium vapor to form. The conceptual continuous model is

$$\frac{Dx}{Dt} = \frac{x_{eq} - x}{\tau}, \quad (33)$$

where τ is a fitted relaxation time. The implementation uses the exact update for constant x_{eq} over one time step:

$$x_{j+1} = x_j + \left(1 - e^{-\Delta t_j/\tau}\right) (x_{eq,j+1} - x_j). \quad (34)$$

9.2 Nucleation delay

Relaxation is allowed only below

$$P_{nuc} = \max(0, P_{bubble} - \Delta P_{delay}). \quad (35)$$

A larger ΔP_{delay} lets liquid remain metastable to lower pressure. Equation (35) is at `src/fuelnozzle/lng.py:385–389`.

9.3 Residence time

Pressure samples are mapped to equally spaced axial positions. For segment length Δz and average endpoint speed,

$$\bar{u}_j = \max\left(0.5, \frac{u_j + u_{j+1}}{2}\right), \quad \Delta t_j = \frac{\Delta z}{\bar{u}_j}. \quad (36)$$

The floor of 0.5 m/s prevents division by zero at the inlet. Implemented at `src/fuelnozzle/lng.py:405–408`. Equation (34) is at lines 410–416.

9.4 Actual mixture density

For $x > 0$, the code uses the homogeneous density from Equation (21). For $x = 0$, it retains compressed-liquid inlet density. Implemented by `_relaxed_mixture_density` at `src/fuelnozzle/lng.py:539–552`.

9.5 Bounded mass flux

The direct estimate is $G_{raw} = \rho_{actual} u_{HEM}$. The implementation prevents this reduced model from leaving the HEM/SPI bracket:

$$G_{Tier2} = \min[G_{SPI}, \max(G_{HEM}, G_{raw})]. \quad (37)$$

Implemented at `src/fuelnozzle/lng.py:428–446`.

Engineering caution. Tier 2 does not solve coupled mass, momentum, energy, and phase-relaxation equations. It relaxes quality along the Tier 1 pressure/velocity path and applies a physical bracket. It should be called a finite-rate screening model, not a validated full HRM solver.

10 LNG Tier 3: Spray Regime and Calibration

10.1 Pressure ratio, superheat, and Jakob number

The saturation-to-chamber pressure ratio is

$$R_p = \frac{P_{bubble}(T_{in}, \mathbf{z})}{P_3}. \quad (38)$$

Superheat relative to the bubble temperature at P_3 is

$$\Delta T_{sh} = \max(0, T_{in} - T_{bubble}(P_3, \mathbf{z})). \quad (39)$$

Latent heat estimate and Jakob number are

$$h_{fg} = h_{dew}(P_3) - h_{bubble}(P_3), \quad (40)$$

$$\text{Ja} = \frac{c_p \Delta T_{sh}}{h_{fg}}. \quad (41)$$

Implemented by `solve_lng_flash_spray` at `src/fuelnozzle/spray.py:102`; equations occur at lines 126–143.

10.2 Regime classifier

`_classify_regime` begins at `src/fuelnozzle/spray.py:225`. The classifier first honors upstream, no-flash, and external-flash results from lower tiers. Remaining cases are fully flashing when any of these default conditions is true:

$$x_{exit} \geq 0.10 \quad \text{or} \quad R_p \geq 3.0 \quad \text{or} \quad \text{Ja} \geq 0.10. \quad (42)$$

Otherwise the case is transitional. These thresholds are exposed in `FlashSpraySettings` at `src/fuelnozzle/spray.py:35`; they are not universal physical boundaries.

10.3 Calibrated LNG SMD

No SMD is produced without calibration. With a reference SMD, pressure ratio, and flash quality, the code uses

$$D_{32} = D_{32,ref} \left(\frac{R_p}{R_{p,ref}} \right)^{-a} \left(\frac{x_{drive}}{x_{ref}} \right)^{-b}, \quad (43)$$

where

$$x_{drive} = \max(10^{-6}, x_{exit}, x_{eq,P_3}). \quad (44)$$

The uncertainty interval is

$$[D_{32}(1 - U_D), D_{32}(1 + U_D)]. \quad (45)$$

Implemented at `src/fuelnozzle/spray.py:168–181`.

10.4 Calibrated cone angle

The full cone-angle estimate is

$$\Theta = \text{clip}_{[0,180]} \left[\Theta_{ref} + K_{\Theta} \ln \left(\frac{R_p}{R_{p,ref}} \right) \right]. \quad (46)$$

The reported range adds and subtracts a calibration uncertainty and clips it to $[0, 180]$ degrees. Implemented at `src/fuelnozzle/spray.py:182–189`.

Engineering caution. Equations (43) and (46) are calibration forms selected for transparent screening. They are not universal flash-atomization correlations.

11 Jet-A Pressure-Swirl Model

11.1 Jet-A properties

Jet-A is a variable mixture rather than one pure chemical. The code therefore accepts density, viscosity, surface tension, and optional vapor pressure from a measured or declared source. `JetAPr` `opertyTable.at_temperature` linearly interpolates a monotonic temperature table at `src/fuelnozzle/jet_a.py:64`. Outside the table it uses an endpoint and emits a warning.

11.2 Ideal speed and exit sizing

The Bernoulli ideal speed is

$$u_{ideal} = \sqrt{\frac{2\Delta P_n}{\rho}}. \quad (47)$$

The orifice relation is

$$\dot{m} = C_d \rho A_o u_{ideal} = C_d A_o \sqrt{2\rho\Delta P_n}. \quad (48)$$

Solving for area and diameter gives

$$A_{req} = \frac{\dot{m}}{C_d \rho u_{ideal}}, \quad (49)$$

$$D_{req} = \sqrt{\frac{4A_{req}}{\pi}}. \quad (50)$$

Implemented by `solve_jet_a_pressure_swirl` at `src/fuelnozzle/jet_a.py:143`; equations occur at lines 168–180.

11.3 Tangential port velocity and angular momentum

For N_p circular ports of diameter D_p ,

$$A_p = N_p \frac{\pi D_p^2}{4}, \quad u_p = \frac{\dot{m}}{\rho A_p}. \quad (51)$$

A total exit speed is formed using velocity coefficient C_v :

$$u_{tot} = C_v u_{ideal}. \quad (52)$$

The unconstrained tangential speed follows a simple angular-momentum transfer:

$$u_{\theta,raw} = \eta_L u_p \frac{r_t}{r_o}, \quad (53)$$

where r_t is port tangency radius, r_o is exit radius, and η_L is an empirical efficiency. Implemented at `src/fuelnozzle/jet_a.py:183–193`.

The code caps tangential speed using total pressure energy and a minimum axial speed needed for continuity. Axial speed is then

$$u_z = \sqrt{\max(10^{-12}, u_{tot}^2 - u_{\theta}^2)}. \quad (54)$$

Implemented at `src/fuelnozzle/jet_a.py:194–217`.

11.4 Air core and film thickness

The liquid annular area required by continuity is

$$A_l = \frac{\dot{m}}{\rho u_z}. \quad (55)$$

For exit radius r_o , the air-core radius is

$$r_a = \sqrt{r_o^2 - \frac{A_l}{\pi}}, \quad (56)$$

and film thickness is

$$t_f = r_o - r_a. \quad (57)$$

Implemented at `src/fuelnozzle/jet_a.py:218–234`. If $A_l \geq A_o$, the code sets no air core and returns an error warning.

11.5 Cone angle

The nominal full cone angle is twice the velocity-vector semi-angle:

$$\Theta = 2 \tan^{-1} \left(\frac{u_{\theta}}{u_z} \right). \quad (58)$$

Implemented at `src/fuelnozzle/jet_a.py:236`. This is simpler than a complete Abramovich/Giffen-Muraszewski internal-flow solution.

11.6 Cavitation screen

Centrifugal velocity is assumed to depress internal pressure by dynamic pressure:

$$P_{min} = P_{n,in} - \frac{1}{2} \rho u_{\theta}^2. \quad (59)$$

The vapor-pressure margin is

$$\Delta P_{cav} = P_{min} - P_{vap}(T). \quad (60)$$

Implemented at `src/fuelnozzle/jet_a.py:237–244`. Negative margin triggers an error-level warning. This is a screen, not a resolved cavitation model.

11.7 Dimensionless atomization groups

Using film thickness as length scale,

$$\text{Re}_f = \frac{\rho u_{tot} t_f}{\mu}, \quad (61)$$

$$\text{We}_f = \frac{\rho u_{tot}^2 t_f}{\sigma}, \quad (62)$$

$$\text{Oh}_f = \frac{\mu}{\sqrt{\rho \sigma t_f}}. \quad (63)$$

Implemented at `src/fuelnozzle/jet_a.py:262–275`. Reynolds number compares inertia to viscosity, Weber number compares inertia to surface tension, and Ohnesorge number compares viscosity to inertial-capillary forces.

Ambient air density is estimated as an ideal gas:

$$\rho_a = \frac{P_3}{R_a T_3}, \quad R_a = 287.05 \text{ J/(kg K)}. \quad (64)$$

Implemented at `src/fuelnozzle/jet_a.py:276–278`.

11.8 Calibrated Jet-A SMD

The code defines a capillary sheet scale

$$\ell_c = \sqrt{\frac{\sigma t_f}{\rho u_{tot}^2}} \quad (65)$$

and calibrated SMD

$$D_{32} = C_{SMD} \ell_c (1 + 3\text{Oh}_f). \quad (66)$$

Implemented at `src/fuelnozzle/jet_a.py:294–303`. No SMD is returned unless C_{SMD} is supplied.

Engineering caution. The current geometry fields for swirl-chamber radius, swirl-chamber length, and exit-orifice length are validated inputs but do not enter the hydraulic equations. A future named pressure-swirl model must use them or remove them from the claimed physics.

12 Mission-Level Integration

If every operating point has duration Δt_i and multiplier N_i , integrated fuel mass is

$$m_f = \sum_i \dot{m}_{f,i} \Delta t_i N_i. \quad (67)$$

Implemented by `_integrate_fuel_masses` at `src/fuelnozzle/study.py:170`; Jet-A and LNG sums are lines 175–184.

For external mission total m_{ref} , mismatch is

$$e_{rel} = \frac{|m_f - m_{ref}|}{\max(m_{ref}, 10^{-12})}. \quad (68)$$

Implemented by `_mission_mass_warnings` at `src/fuelnozzle/study.py:188`, equation at line 199. The default warning tolerance is 5%.

13 Complete Function-to-Source Map

Function or class	Source start	Responsibility
<code>OperatingPoint</code>	<code>src/fuelnozzle/operating.py:12</code>	Per-stage user inputs
<code>fuel_pressure_budget</code>	<code>src/fuelnozzle/operating.py:55</code>	Pump/feed/nozzle pressure closure
<code>LNGComposition</code>	<code>src/fuelnozzle/models.py:28</code>	Normalize LNG composition
<code>CoolPropLNGProvider.__init__</code>	<code>src/fuelnozzle/properties.py:29</code>	Reusable EOS state creation
<code>state_pt</code>	<code>src/fuelnozzle/properties.py:60</code>	Pressure-temperature flash
<code>state_ph</code>	<code>src/fuelnozzle/properties.py:72</code>	Pressure-enthalpy flash/root
<code>state_ps</code>	<code>src/fuelnozzle/properties.py:96</code>	Pressure-entropy flash/root
<code>state_pq</code>	<code>src/fuelnozzle/properties.py:120</code>	Pressure-quality saturation state
<code>state_tq</code>	<code>src/fuelnozzle/properties.py:132</code>	Temperature-quality saturation state
<code>_mixture_target_state</code>	<code>src/fuelnozzle/properties.py:152</code>	Bracketed mixture PH/PS root
<code>_component_transport_fallback</code>	<code>src/fuelnozzle/properties.py:281</code>	Approximate missing mixture transport
<code>_mass_quality</code>	<code>src/fuelnozzle/properties.py:370</code>	Convert molar to mass quality
<code>solve_lng_feed_line</code>	<code>src/fuelnozzle/feed.py:63</code>	Segmented cryogenic feed solver
<code>_heat_leak_per_length</code>	<code>src/fuelnozzle/feed.py:230</code>	Measured or calculated heat ingress
<code>_liquid_segment_pressure_drop</code>	<code>src/fuelnozzle/feed.py:270</code>	Single-phase friction/minor/static loss
<code>_two_phase_segment_pressure_drop</code>	<code>src/fuelnozzle/feed.py:297</code>	ChEDL two-phase friction and static loss
<code>screen_lng_flash</code>	<code>src/fuelnozzle/lng.py:151</code>	Tier 0 equilibrium phase screen
<code>solve_lng_equilibrium_flow</code>	<code>src/fuelnozzle/lng.py:256</code>	Tier 1 SPI/HEM sizing and choking
<code>solve_lng_relaxation_flow</code>	<code>src/fuelnozzle/lng.py:363</code>	Tier 2 finite-rate flashing screen
<code>_relaxed_mixture_density</code>	<code>src/fuelnozzle/lng.py:539</code>	Homogeneous actual-quality density
<code>solve_lng_flash_spray</code>	<code>src/fuelnozzle/spray.py:102</code>	Tier 3 spray outputs and calibration

Function or class	Source start	Responsibility
<code>_classify_regime</code>	<code>src/fuelnozzle/spray.py:225</code>	Rule-based flash-spray class
<code>JetAPropertyTable.at_temperature</code>	<code>src/fuelnozzle/jet_a.py:64</code>	Jet-A property interpolation
<code>solve_jet_a_pressure_swirl</code>	<code>src/fuelnozzle/jet_a.py:143</code>	Jet-A hydraulic and spray screen
<code>run_nozzle_study</code>	<code>src/fuelnozzle/study.py:75</code>	Whole-envelope orchestration
<code>_integrate_fuel_masses</code>	<code>src/fuelnozzle/study.py:170</code>	Stage-to-mission fuel integration
<code>_mission_mass_warnings</code>	<code>src/fuelnozzle/study.py:188</code>	External mission consistency check

14 Existing Verification Coverage

The current suite has 21 tests. The main test starts are:

- properties: `tests/test_properties.py:7`, 19, and 51;
- feed line: `tests/test_feed.py:31` and 48;
- LNG Tier 0/1: `tests/test_lng_equilibrium.py:37`, 50, 63, and 81;
- LNG Tier 2: `tests/test_lng_relaxation.py:36`, 52, and 72;
- LNG Tier 3: `tests/test_spray.py:35`, 54, and 80;
- Jet-A: `tests/test_jet_a.py:41`, 67, and 83;
- full study: `tests/test_study.py:47`, 70, and 86.

These tests confirm code paths, round trips, bounds, warning behavior, and basic limiting cases. Required next verification includes pressure-grid convergence, Tier 2 axial-grid convergence, independent analytical values, energy/momentum residuals, and property tables.

15 Validation Evidence and Planned Use

15.1 Critical flow

Hammer et al. [3] provide modern CO₂ orifice/nozzle choking data. CO₂ is not an LNG surrogate; it tests the HEM and non-equilibrium mathematical structure for a strongly real fluid.

De Lorenzo et al. [4] benchmark delayed-equilibrium and classic critical-flow models against multiple water experiments, including Super Moby Dick. This is the primary source for testing relaxation-model behavior.

Kim and O’Neal [5] and the Payne/O’Neal compilation provide short-orifice refrigerant evidence for metastability and geometry dependence.

15.2 Cryogenic spray

DaRUS-2527 [6] contains 22 LN2 shadowgraph cases and condition data. It is the preferred Tier 3 regime/angle dataset. Rees et al. [7] provide interpretation.

DaRUS-2076 [8] and Rees et al. [9] provide raw local D_{10} and velocity data. These validate cryogenic spray fields but do not directly validate LNG D_{32} .

Gaertner et al. [10] provide a numerical/experimental cryogenic comparison. The correct article number is 103360.

15.3 Pressure swirl and Jet-A

Lacava et al. [11] provide a complete, open, four-port water atomizer benchmark. At the 4-atm design point the reported experimental values are $\dot{m} = 6.13$ g/s, $C_d = 0.2728$, semi-angle 34.5° , and $D_{32} = 80.83 \mu\text{m}$. Water data verify model form, not Jet-A transfer.

Dodge and Biaglow [12] provide direct Jet-A SMD and distribution trends with elevated air pressure and temperature. The NASA record is metadata-only, so a full paper must be transcribed before quantitative tests are created.

16 Known Deviations and Technical Debt

1. P_3 is treated as nozzle back pressure; no separate liner pressure loss exists.
2. The feed model is steady and uniform. It omits pump dynamics, valve transients, conjugate transient wall storage, and a separate two-phase acceleration loss.
3. Tier 0 onset position assumes a linear pressure profile.
4. Tier 1 samples an isentropic thermodynamic path; it does not solve an area/friction-coupled nozzle differential equation.
5. Choking is a sampled mass-flux maximum rather than an explicit sonic condition.
6. Tier 2 is a quality-relaxation screen on the Tier 1 path, not a full HRM PDE/ODE conservation solver.
7. Tier 2 mass flux is clamped to HEM/SPI bounds. This is a guardrail, not a derived non-equilibrium critical-flow law.
8. Tier 3 regime thresholds and calibrated SMD/cone equations require validation.
9. LNG transport mixing is approximate where CoolProp has no native mixture model.
10. Jet-A swirl chamber radius, chamber length, and exit length are not yet used by the hydraulic closure.
11. The Jet-A cone angle uses a velocity-vector approximation rather than a complete pressure-swirl theory.
12. No automated parameter uncertainty propagation is implemented.
13. No direct adapter to the separate mission repository is implemented; only generic operating points and mission-total checks are supported.
14. CFD export is an in-memory boundary record, not a solver-specific file writer.

17 Recommended Next Steps

1. Add numerical-convergence tests before fitting any empirical parameter.
2. Create immutable raw-data and derived-data folders with source checksums.
3. Transcribe one Hammer CO₂ orifice series and all Lacava curves first.
4. Generalize the property protocol so the same Tier 1 kernel can run CO₂, R134a, water, nitrogen, methane, and LNG without changing equations.
5. Fit Tier 2 on declared training cases and evaluate held-out geometries/states.
6. Load DaRUS datasets directly rather than digitizing available CSV/XLSX values.
7. Add named pressure-swirl correlation adapters and use currently unused geometry.
8. Add direct LCH₄/LNG validation only when composition, inlet state, geometry, back pressure, mass flow, and uncertainty are all available.

18 Building This Report

The project uses Tectonic through Pixi:

```
pixi install
pixi run doc
```

The task is defined in `pixi.toml`. Tectonic downloads required TeX packages and writes the PDF next to this source file.

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